Lecture 2, Dec 18th, 2025.

Properties: i). $|a \cdot b| = |a| \cdot |b|$

ii). $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$ if $b \neq 0$.

iii). $|a^n| = |a|^n$

Now assume $a > 0$ then.

i). $|x| = a$ if and only if $x = \pm a$. ✓

ii). $|x| < a$ if and only if $-a < x < a$. ($-a < x$ and $x < a$) ✓

iii). $|x| > a$ if and only if $x > a$ or $x < -a$

Problem. Solve $x^3 + 3x^2 > 4x$.

Sln: **Step 1.** Unequal to zero.

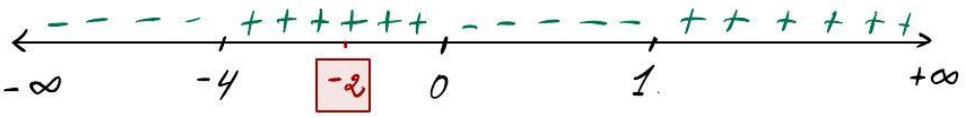
$$x^3 + 3x^2 - 4x > 0$$

Step 2. Factorize.

$$x(x^2 + 3x - 4) > 0$$

$$x(x+4)(x-1) > 0$$

Step 3. Draw a line with the roots of the polynomial above.



Step 4. Select "easy" number and evaluate the sign

if $x = -2$ $\quad (-2)(-2+4)(-2-1) = (-2)(2)(-3) = +12$.

Step 5. Select the region for your problem.

$x(x+4)(x-1) > 0$ (positive numbers)

$$S = (-4, 0) \cup (1, +\infty).$$

Example. iii). Solve $|3x+2| \geq 4 \overset{\text{def}}{\iff}$ $3x+2 \geq 4$ or $3x+2 \leq -4$

$3x \geq 4-2$ $\qquad$ $3x \leq -4-2$

$3x \geq 2$ $\qquad$ $3x \leq -6$

$x \geq \frac{2}{3}$ $\qquad$ $x \leq -\frac{6}{3}$

$x \leq -2.$

$S_1 = [\frac{2}{3}, \infty)$ $\qquad$ $S_2 = (-\infty, -2]$

then

$$S = S_1 \cup S_2 = (-\infty, -2] \cup [\frac{2}{3}, \infty)$$

Problem. Solve $\frac{1}{x} < 4$.

Sln:

$$\frac{1}{x} - \frac{4}{1} < 0.$$

$$\frac{1-4x}{x} < 0$$

$\frac{1-4(2)}{2} = \frac{1-8}{2} = \frac{-7}{2}$



$$S = (-\infty, 0) \cup (\tfrac{1}{4}, \infty).$$

Problems. i). Prove the triangular inequality.

For all $a, b \in \mathbb{R}$

$$|a+b| \leq |a| + |b|.$$

ii) Solve for x, assuming that a, b, and c are negative constants.

$$ax + b < c.$$

Sln:

i). Let $a, b \in \mathbb{R}$ and notice that

$$-|a| \leq a \leq |a|.$$

$$-|b| \leq b \leq |b|$$

$$-|a| - |b| \leq a+b \leq |a| + |b|$$

$$-(\underbrace{|a| + |b|}_{-s}) \leq \underbrace{a+b}_{x} \leq \underbrace{|a| + |b|}_{s}$$

by definition $\qquad |a+b| \leq |a| + |b|.$

ii). $ax + b < c$ Solving for x.

$ax < c - b$.

$x > \frac{c-b}{a}$ b/c a is negative number.

$S = \left(\frac{c-b}{a}, \infty\right)$.

Limit of a function (Intuitive).

what is the result of $\frac{0}{0}$?

$\frac{1}{1} = 1 \quad \frac{2}{2} = 1 \quad \frac{10}{10} = 1$

If you can try define $\frac{0}{0} = 1$ then

$1 = \frac{0}{0} = \frac{2 \times \cancel{0}}{\cancel{0}} = 2$. Since $1 \neq 2$ then, there is not correct definition of $\frac{0}{0}$, that is, $\frac{0}{0}$ is not a number.

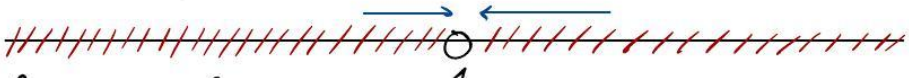
Sometimes the functions present this problem.

Let $f(x) = \frac{x-1}{x^2-1}$.

Notice that if $x \neq 1$ then $f(x) = \frac{(x-1)}{(x-1)(x+1)} = \frac{1}{x+1}$

if $x = 1$, what is $f(1)$?

Notice that is not possible compute $f(1)$, but we can ask yourself about what is "approximately" $f(1)$.



For values of x near to 1 we have.

$x < 1$	$f(x)$	$x > 1$	$f(x)$
0.5	0.666667	1.5	0.400000
0.9	0.526316	1.1	0.476190
0.99	0.502513	1.01	0.497512
0.999	0.500250	1.001	0.499750
0.9999	0.500025	1.0001	0.499975
↓	↓	↓	↓
1	0.5	1	0.5

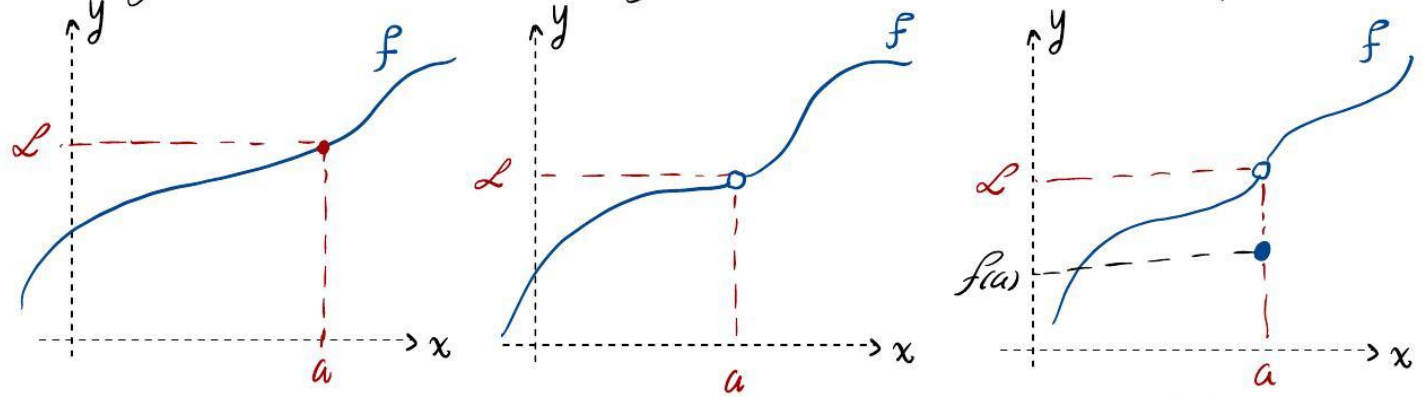
We see that the closer x is to 1 (on either sides of 1) the closer $f(x)$ is 0.5. In fact, it appears that we can make values of $f(x)$ as close as we like to 0.5 by taking x sufficiently close to 1.
We express this by saying "the limit of the function $f(x)=\frac{x-1}{x^2-1}$ as x approaches 1 is equal to 0.5" or.

$$\lim_{x\to 1} \frac{x-1}{x^2-1} = 0.5$$

Def. (Intuitive Definition of Limit) Suppose that the function f is defined on some open interval that contains a, x (except possibly at a itself) the write

$$\lim_{x\to a} f(x) = L$$

If we can make the values of $f(x)$ arbitrarily close to L by restricting x to be sufficiently close to a but not equal to a.



In (I) $\lim_{x\to a} f(x) = L$ and $f(a) = L$.

In (II) $\lim_{x\to a} f(x) = L$ but $f(a)$ is not defined.

In (III) $\lim_{x\to a} f(x) = L$, $f(a)$ is defined but $f(a) \neq L$

Example. Estimate the value of $\lim_{t\to 0} \dfrac{\sqrt{t^2+9}-3}{t^2}$.

Sln: if $t=\pm 1$ $\quad f(\pm 1)=\dfrac{\sqrt{1+9}-3}{1}=\sqrt{10}-3\approx 0.1622.$

if $t=\pm\frac{1}{2}$ $\quad f(\pm\frac{1}{2})=\dfrac{\sqrt{\frac{1}{4}+9}-3}{\frac{1}{4}}=\dfrac{\sqrt{\frac{37}{4}}-3}{\frac{1}{4}}\approx 0.1655.$

$\vdots$

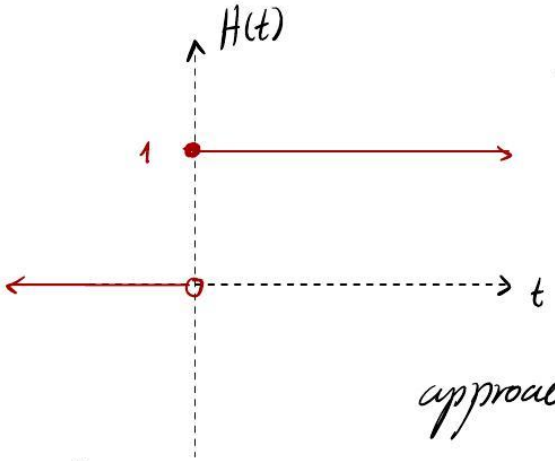
if $t=\pm\frac{1}{10}$ then $f(\pm\frac{1}{10})=\dfrac{\sqrt{1/100+9}-3}{1/100}\approx 0.1666..$

So, when $t\to 0$ then $f(x)\longrightarrow 0.1666\ldots=\frac{1}{6}$.

In the next class we can check this using the rules.

Example. (for you) Guess the value of $\lim_{x\to 0}\dfrac{\sin x}{x}$.

One-sided Limits Let consider



$$\mathcal{H}(t)=\begin{cases}0 & \text{if } t<0\\ 1 & \text{if } t\geq 0.\end{cases}$$

$\lim_{t\to 0} H(t)$ does not exist because there is not a single value that $H(t)$ approaches as t asproaches to 0.

However, as t approaches to 0 from the left $H(t)$ approaches to 0. and as t approaches to 0 from the right $H(t)$ approaches to 1.

So, we can write this

$\lim_{t\to 0^-} \mathcal{H}(t)=0$ (left side) and $\lim_{t\to 0^+} \mathcal{H}(t)=1$ (right side).

Def (Intuitive Definition of one-sided limits) We write

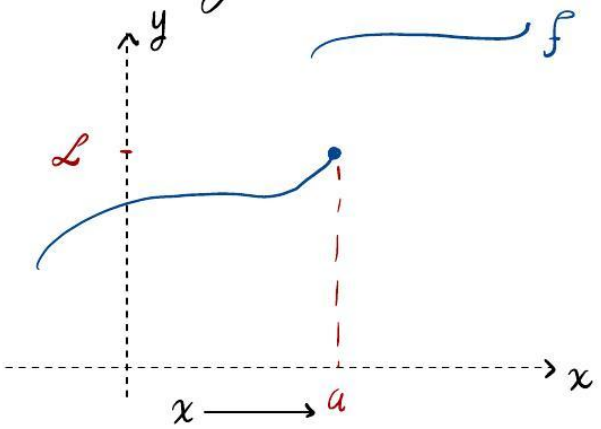
$$\lim_{x \to a^-} f(x) = L.$$

and say that the left-hand limit of $f(x)$ as $x \to a$ is equal to L, if we can make the values of $f(x)$ arbitrary close to L by restricting x to be sufficiently close to a with $x < a$.

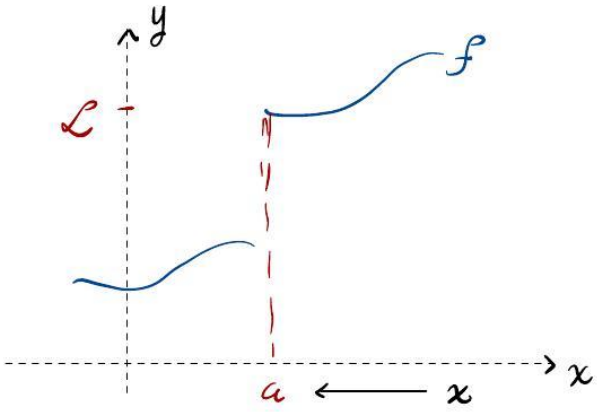
Similarly, we write

$$\lim_{x \to a^+} f(x) = L$$

and call it right-hand limit.



$$\lim_{x \to a^-} f(x) = L.$$

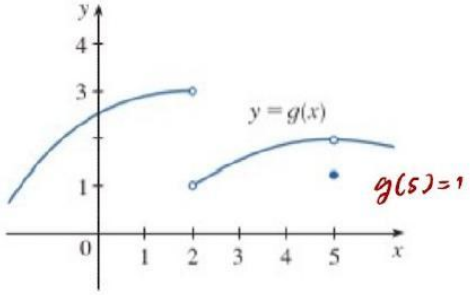


$$\lim_{x \to a^+} f(x) = L.$$

Property ♡

$\lim_{x \to a} f(x) = L$ if and only if $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L$.

Example:



Compute:

a). $\lim_{x \to 2^-} g(x) = 3.$

b). $\lim_{x \to 2^+} g(x) = 1$

c). $\lim_{x \to 2} g(x) =$ DNE

d). $\lim_{x \to 5^-} g(x) = 2$

e). $\lim_{x \to 5^+} g(x) = 2$

f). $\lim_{x \to 5} g(x) = 2$

Example. Find $\lim_{x \to 0} \frac{1}{x^2}$ if it exists.

Sln. As x becomes close to 0, x^2 becomes close to 0, and $1/x^2$ becomes very large.

For example

if $x = \pm \frac{1}{2}$ then $f(\pm \frac{1}{2}) = \frac{1}{\frac{1}{4}} = 4$

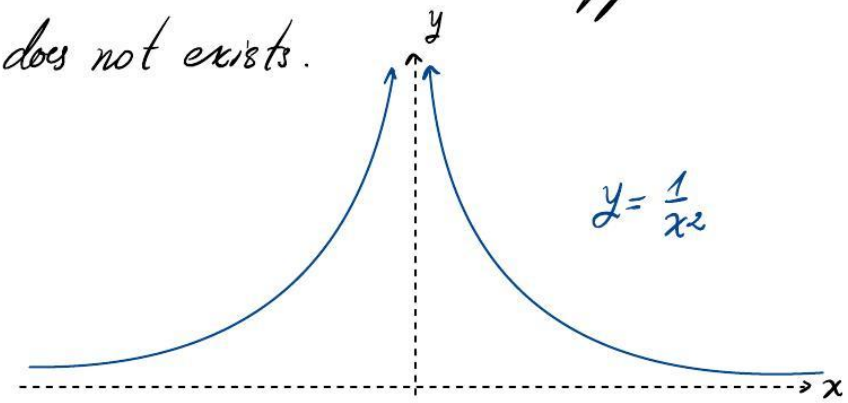
if $x = \pm \frac{1}{5}$ then $f(\pm \frac{1}{5}) = \frac{1}{\frac{1}{25}} = 25$

if $x = \pm \frac{1}{10}$ then $f(\pm \frac{1}{10}) = \frac{1}{\frac{1}{100}} = 100$

⋮

if $x = \pm 0.0001$ then $f(\pm \frac{1}{1000}) = 1.000.000$.

In fact, the values of $f(x)$ can be made arbitrary large by taking x close to 0. Then $f(x)$ do not approach a number, so $\lim_{x \to 0} \frac{1}{x^2}$ does not exists.

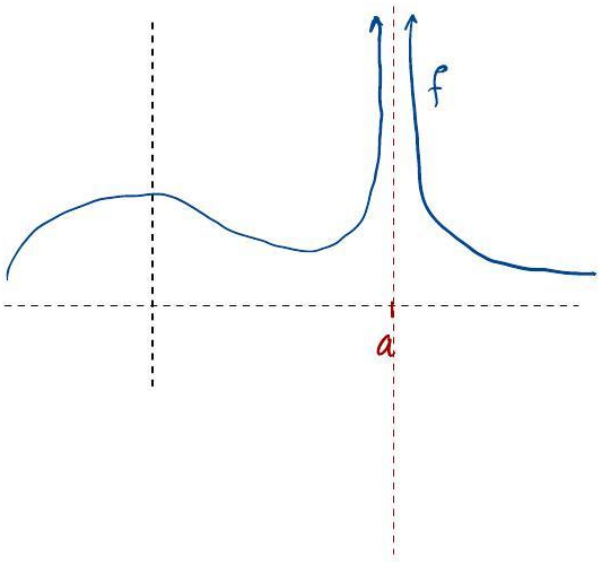


Infinite Limits; Vertical Asymptotes.

To indicate the kind of behavior exhibited in above example we use the notation $\lim_{x \to 0} \frac{1}{x^2} = \infty$

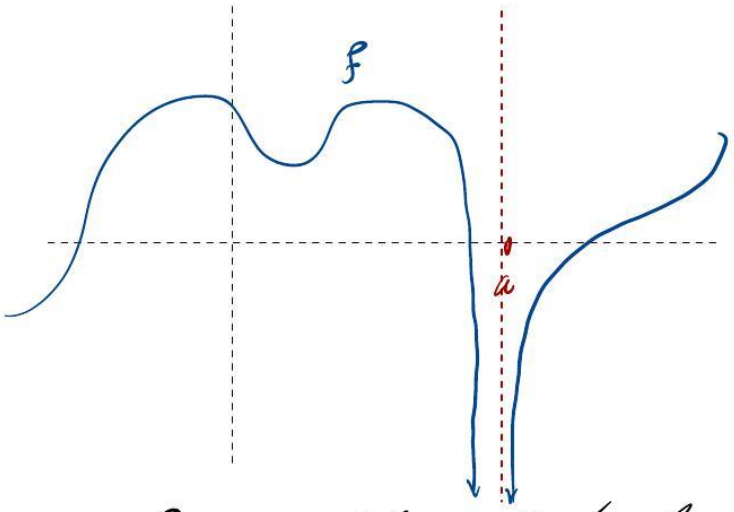
Def. (Intuitive Definition of an infinite limit) Let f be a function defined on both sides of a, except possibly at a itself. Then $\lim_{x \to a} f(x) = \infty$ means that the values of $f(x)$ can be made large

by taking x values close to a, but not equal to a.



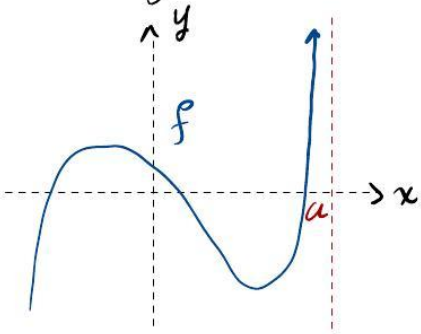
$$\lim_{x \to a} f(x) = \infty.$$

Similarly, for function as

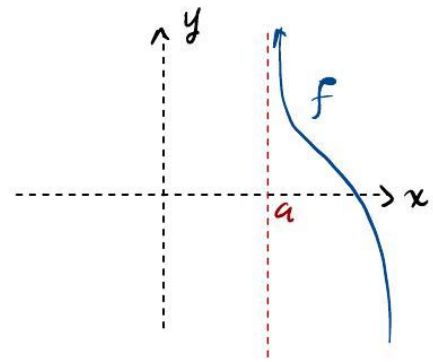


$$\lim_{x \to a} f(x) = -\infty.$$

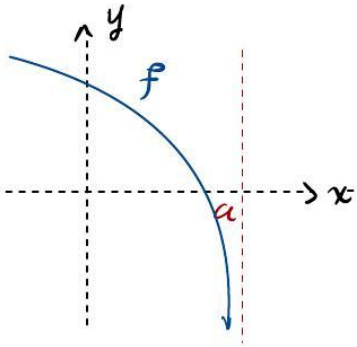
Similarly definitions can be given for one-sided infinite limits.



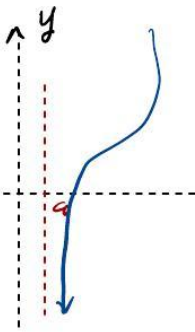
$$\lim_{x \to a^-} f(x) = \infty$$



$$\lim_{x \to a^+} f(x) = \infty$$



$$\lim_{x \to a^-} f(x) = -\infty$$



$$\lim_{x \to a^+} f(x) = -\infty$$

Def. the vertical line $x = a$ is called a vertical asymtote of the curve $y = f(x)$ if at least one of the following statements is true.

$$\lim_{x \to a} f(x) = \infty \qquad \lim_{x \to a^-} f(x) = \infty \qquad \lim_{x \to a^+} f(x) = \infty$$

$$\lim_{x \to a} f(x) = -\infty \qquad \lim_{x \to a^-} f(x) = -\infty \qquad \lim_{x \to a^+} f(x) = -\infty.$$

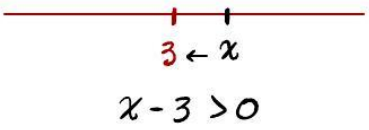
Example. Does the curve $y = \frac{2x}{x-3}$ have a vertical asymptote?

In other words is there any number a s.t

$$\lim_{x \to a} f(x) = \pm\infty.$$

Soln. There is a potential vertical asymtote where the denominador is 0, that is at $x = 3$. So, we investigate the one-sided limits here.

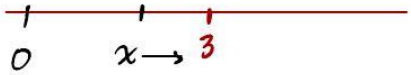
$$\lim_{x \to 3^+} \frac{2x}{x-3} = +\infty$$



$x - 3 > 0$

if $x > 3$ then $x - 3 > 0$ and $2x > 0$ then $\frac{2x}{x-3} > 0$

$$\lim_{x \to 3^-} \frac{2x}{x-3} = -\infty$$



if $x < 3$ then $x - 3 < 0$ and $x > 0$ b/c x is close to 3 then $2x > 0$ so $\frac{2x}{x-3} < 0$